

Supplementary Material for CIL: Content-Identity Auditing of Temporal Score Curves

Author Names Required Before Submission

I. FROZEN DECISION RULES

For each target video, target-identity advantage is the temporal average precision of its own tie-aware percentile-rank curve minus the mean average precision of eight deterministic progress-normalized donor curves on the same target labels. Content-necessity margin subtracts the maximum identity advantage of the frozen sentinel family within every bootstrap draw. All primary intervals resample target videos under the fixed donor schedule rather than treating frames as independent.

For a curve of length T , the rank of a tied group is its mean one-based ordinal rank a_t , and the percentile value is $(a_t - 0.5)/T$. Donor ranks are linearly interpolated on normalized progress with endpoint extension where required. AP groups equal scores at a common threshold: if P_k and R_k are precision and recall at successive distinct descending score levels, then $A = \sum_k (R_k - R_{k-1})P_k$. Mixed-label targets alone enter the ownership mean; all-positive AP is mathematically one but supplies no ranking contrast, whereas all-normal targets have no positive class. Thus exclusion of both constant label types defines the estimand rather than a search for favorable effects.

The Pistachio protocol uses 10,000 draws, seed 17082026, five hash-ordered strata, and donor shifts 7, 19, 31, 43, 59, 73, 91, and 109. The IITB protocol independently uses the same shifts and sentinel names, seed 24082026, and maximum one-sided 95% simultaneous lower bounds over all three candidates. IITB additionally requires at least 100 donors, 60 evaluable targets, TIA at least .02, CNM at least .015, positive marginal and simultaneous lower bounds, at least four nonnegative strata, frame AUC at least .60, and AP at least prevalence plus .05.

A. Why these donor shifts and effect floors?

The values 7, 19, 31, 43, 59, 73, 91, and 109 are offsets in a SHA-256 hash-order of video identities, not temporal lags. They were chosen before the corresponding confirmation to provide eight distinct, nonzero, broadly spaced owners without using scores or labels. Each role reduces them modulo its donor count and rejects a zero or duplicate effective shift. Thus every target has the same donor count, no curve donates to itself, and the rule is replayable. The number eight is a fixed coverage/compute compromise, not an outcome-tuned hyperparameter or a claim that eight donors exhaust the population.

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The point floors prevent a numerically tiny but precisely estimated contrast from being promoted. They are operational minimum-effect thresholds in video-macro AP units, not natural constants and not comparable across datasets as detector scores. The OOPS source/confirmation TIA floors are .035/.030 and CNM floors are .030/.025; the source role is intentionally stricter than the untouched confirmation. Pistachio and IITB each use TIA .020 and CNM .015. These values were locked before their corresponding scoring or label opening; the artifacts do not support a stronger claim that the numbers are uniquely optimal. Positive marginal intervals, fixed strata, replay, and label-flow checks are required in addition to the floors.

B. Train-only Pistachio scores and analytic controls

Each 512-dimensional ViT segment feature is individually ℓ_2 -normalized with a 10^{-12} denominator floor. Segment supervision is inherited from the official training video's normal/anomalous class. The centroid direction is the normalized difference between anomaly and normal training means. For ridge, features are augmented with an intercept and each class receives total weight one half. With balanced second-moment matrix G and balanced signed mean g , the parameter is $(G + \Lambda)^{-1}g$, where feature penalties equal $10^{-3} \text{tr}(G_{1:512,1:512})/512$ and the intercept is unpenalized. Neither direction nor regularization is selected using confirmation outcomes.

For Pistachio and IITB, the four analytic controls on their respective construction grids of length L use $u_t = t/(L - 1)$: a Gaussian center prior of width .22, a Gaussian center prior of width $\min(.45, \max(.12, 8/L))$, linear progress u_t , and $\sin^2(\pi(t + .5)/4)$. Pistachio constructs them on the feature grid before transfer; IITB constructs them on the aligned evaluation grid as specified below. These fixed grids are part of the estimand, not interchangeable implementations of the same continuous curve.

II. COMPLETE PROSPECTIVE ENDPOINT RECORD

IITB pools all 150 test videos for 181,567 aligned frames at prevalence .59635. Normal distance has AUC .68184 and AP .72902. Temporal novelty has AUC .71640 and AP .75231; frame difference has AUC .66561 and AP .71020. Their simultaneous one-sided TIA/CNM lower-bound pairs are .01562/.01474, .04282/.04194, and .02278/.02190. Every frozen family member passes every scientific and paper-promotion check.

TABLE IV
COMPLETE LOCKED SOURCE AND CONFIRMATION ENDPOINT RECORD.

Endpoint	n	TIA [95% CI]	CNM [95% CI]	Strata	Interpretation
OOPS latent motion, source	128	.06675 [.03207,.10217]	.06422 [.03074,.09883]	4/5	qualified
OOPS latent motion, confirmation	256	.05830 [.03304,.08365]	.05751 [.03187,.08304]	5/5	licensed
Pistachio centroid, source	65	.08604 [.04665,.12513]	.08211 [.03815,.12120]	5/5	qualified
Pistachio centroid, confirmation	120	.10051 [.06706,.13486]	.09364 [.05831,.12917]	5/5	licensed
Pistachio ridge, source	65	.11223 [.07610,.14821]	.10830 [.06592,.14498]	5/5	qualified
Pistachio ridge, confirmation	120	.11813 [.08995,.14648]	.11125 [.08093,.14134]	5/5	licensed
IITB normal distance	134	.03879 [.01584,.06201]	.03791 [.01493,.06108]	5/5	licensed + utility
IITB temporal novelty	134	.06599 [.04253,.09094]	.06511 [.04167,.08978]	5/5	licensed + utility
IITB frame difference	134	.04595 [.02523,.06776]	.04507 [.02439,.06661]	5/5	licensed + utility

The 150 IITB label vectors comprise 134 mixed-label, 10 all-normal, and 6 all-anomalous videos. The 16 constant-label videos cannot identify the two-class within-target ordering required by TIA, so they are excluded as targets but retained in the label-independent donor pool. The resulting audit uses 134 targets, 150 unique potential donors, and eight donors per target.

OOPS and Pistachio show “–” for AUC/AP in the main prospective table because ordinary frame utility was not a frozen promotion gate for those historical endpoints, not because utility was assumed. OOPS was contracted as transition-localization ownership using video-macro temporal AP; retroactively adding pooled AUC/AP would change the endpoint after label access. Pistachio does have post-confirmation descriptive raw-frame measurements: on the confirmation role, ridge AUC/AP is .74342/.58263 and centroid is .67311/.47070. They are context only and were not used to license either prospective endpoint. The Pistachio confirmation role contains 257 videos, including 120 mixed-label ownership targets and 137 constant-label videos. Its pooled descriptive metrics use the full role; ownership averages only the 120 targets. IITB, designed later, prospectively couples CIL to raw global frame AUC/AP gates on all 150 aligned videos.

III. POST-CONFIRMATION CROSSED TARGET/DONOR ROBUSTNESS

The primary prospective intervals resample target videos while treating the frozen donor schedule as part of each target effect. To quantify dependence from donor reuse, we reconstruct the AP differences on the eight frozen donor edges per target and draw independent multinomial weights for target videos and unique donor videos. No unobserved pair is evaluated. For edge effects $d_{ij} = A(r_i, y_i) - A(r_j \rightarrow i, y_i)$ and edge set E , each draw uses

$$\widehat{\text{TIA}}^* = \frac{\sum_{(i,j) \in E} w_i v_j d_{ij}}{\sum_{(i,j) \in E} w_i v_j}. \quad (3)$$

The weights w and v are independently drawn multinomial counts over the target and donor roles. They are shared across methods within a draw; empty weighted graphs are discarded. This is an edge-weighted sensitivity analysis, not independent sampling of every possible target–donor pair. The strongest sentinel is recomputed within each draw. OOPS and IITB use 5,000 draws with seeds 26082026 and 26082027; the

previously reported Pistachio analysis uses 5,000 draws with seed 3170026.

The crossed intervals are wider for every endpoint under this resampling scheme. All five marginal lower bounds remain above zero; they are not simultaneous bounds for the added robustness family. Because a recording may occur in both roles and several recordings may share a scene, these intervals do not establish unconditional coverage under arbitrary recording dependence. The sealed OOPS/IITB robustness result has SHA-256

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bc4c6d2652e63a24033ca00eefbcd6bd6
453138d0d910dde277af4a1d7700e92e
```

Its evidence role is explicitly “post-confirmation robustness, not independent confirmation.” It neither changes the original prospective decisions nor adds another dataset.

IV. IITB INFORMATION FIREWALL AND AMENDMENT

Figure 4 separates label-free candidate generation from the label opening and the subsequent disclosed alignment amendment.

The official research-use archives contain 208 training AVI, 150 test AVI, and 150 separately namespaced NPY labels with an exact case-insensitive basename join. The central-directory audit read zero member payload and annotation-body bytes. Media-only extraction produced 358 AVI and zero NPY. Primary and replay prediction ZIPs are byte-identical at SHA-256

```
ef72f5b328b58c51bd75e9f80dfa0f
ea5872af82a6f2ba39794139a5841cc6
```

Only after this seal were 150 NPY labels opened once. All 150 had label length equal to decoded frames minus one and were binary. No scientific metric had been computed. The amendment records the uniform rule “drop score frame zero, then require exact label length” for the complete family. The sealed evaluator implements this literally for all three candidate score arrays. Analytic sentinels, however, are instantiated directly at the final label-vector length; it does not construct native-length sentinels and then trim them. We disclose this implementation distinction rather than claiming numerical equivalence: the duration prior and sampling-grid control can change when their construction grid changes. No label values enter their formulas and the frozen result is

TABLE V

PROSPECTIVE TARGET-VIDEO INTERVALS VERSUS POST-CONFIRMATION CROSSED TARGET/UNIQUE-DONOR INTERVALS. ALL ENTRIES ARE CNM [95% CI]. THE CROSSED COLUMNS ARE ROBUSTNESS CHECKS, NOT NEW CONFIRMATION GATES.

Endpoint	Prospective video-only	Crossed target/donor
OOPS latent motion	.05751 [.03187,.08304]	.05751 [.02592,.08885]
Pistachio ridge	.11125 [.08093,.14134]	.11125 [.07459,.14850]
IITB normal distance	.03791 [.01493,.06108]	.03791 [.00975,.06686]
IITB temporal novelty	.06511 [.04167,.08978]	.06511 [.03667,.09531]
IITB frame difference	.04507 [.02439,.06661]	.04507 [.02050,.07054]

Label-free candidate generation

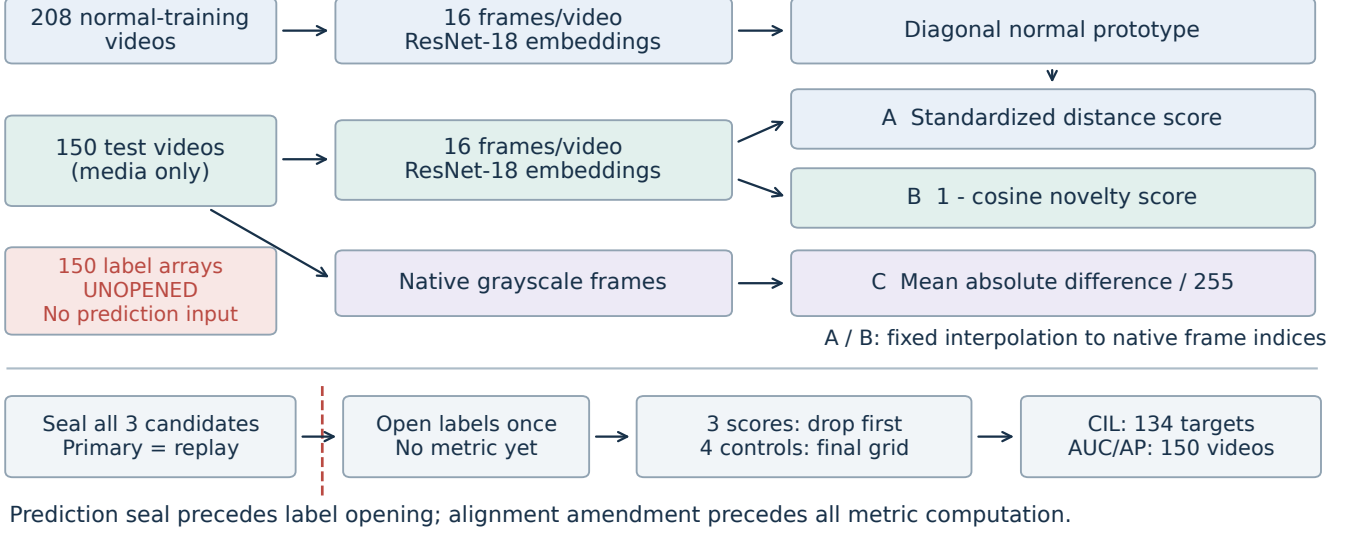


Fig. 4. IITB candidate generation and local information barrier. Training media determine the normal prototype; test media determine the three score curves. Label arrays have no path into prediction generation. After prediction sealing, labels are opened once and the uniform drop-first alignment amendment is recorded before metric computation. Ownership uses 134 mixed-label targets; ordinary utility uses all 150 test videos.

reported without retroactive re-alignment or control replacement. The amendment prohibits drop-last, label padding or interpolation, candidate-specific alignment, and selection by AUC, AP, TIA, or CNM. The result SHA-256 is

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041436e4df223c0a4edaf4827c488efac
b2124bda42130317815e77f49143c2c
```

V. FALSIFIERS, NEGATIVES, AND PRESSURE EVIDENCE

The EgoOops position/length side-information curve attains TIA .10947, exceeding V-JEPA2 ActionFormer TIA .04999 and Omnivore TIA .00032; the corresponding CNM values are negative and both visual candidates are denied. On the consumed Pistachio confirmation role, deterministic wrong-owner reassignment yields -.01085 [-.03693,.01479], while crossed target/donor resampling retains ridge CNM .11125 [.07459,.14850]. Four parameter-free Pistachio C3D/ViT dynamics candidates fail source qualification and remain in the evidence ledger.

VI. ARTIFACT AND CLAIM BOUNDARY

SHA-256 checks bind frozen inputs, claims, and figures; the accompanying tests cover core, IITB, and crossed calculations.

TABLE VI

POST-CONFIRMATION OR RETROSPECTIVE EVIDENCE; NONE IS AN INDEPENDENT CONFIRMATION.

Method/role	AUC/AP	TIA [95% CI] CNM [95% CI]	Status
PEL4VAD source	.72680/.61760	.14331 [.09356,.19339] .13938 [.08673,.18939] .15903 [.12785,.19030]	lic., low util.
PEL4VAD confirm.	.74302/.59244	.15215 [.11998,.18443] .14510 [.11286,.17868]	lic., low util.
Fed-WSVAD source	.78848/.63811	.14117 [.10592,.17412] .12218 [.08975,.15522]	lic., high util.
Fed-WSVAD confirm.	.79198/.61946	.11530 [.08038,.15078] -.00128 [-.01190,.00966]	lic., high util.
Shanghai latent	.44826/.35614	.13776 [.07885,.19898]	denied
Shanghai ConvLSTM	.55032/.48260	.13776 [.07885,.19898] .09988 [.04673,.15466]	CIL, low util.
Shanghai recon.	.46744/.39237	.09988 [.04673,.15466]	CIL, low util.

Restricted media, labels, per-video predictions, credentials, and weights are not redistributed. CIL supports sentinel-relative temporal-score alignment in the audited endpoints, not semantic understanding, physical causality, detector improvement, exhaustive deconfounding, or universal validity.